

F573 — the $\sin^2\theta_W$ ‘ n_C -running’ lead, looked at: it is NOT running — it is a single hypercharge-NORMALIZATION number, and the whole question reduces to ‘is that number $k = \text{rank?}$ ’. Clean structural form: $\sin^2\theta_W = N_c/(N_c + k \cdot n_C)$, where k is the hypercharge normalization. $k=1$ gives the GUT value $3/8 = N_c/(N_c+n_C)$ (the naive embedding, Five-Absence-forbidden); $k=\text{rank}=2$ gives BST’s $3/13 = N_c/(N_c+\text{rank} \cdot n_C)$ (matches observed M_Z 0.2308 vs 0.2312, 0.2%). So the $13-8 = n_C$ shift is NOT a continuous running effect — it is $(\text{rank}-1) \cdot n_C = n_C$, i.e. the hypercharge coupling picking up ONE extra factor of $\text{rank}=2$ in its normalization (equivalently g^2/g'^2 : $5/3 = n_C/N_c \rightarrow 10/3 = \text{rank} \cdot n_C/N_c$). Verified via the coupling ratio: $1/\sin^2\theta_W = 1 + g^2/g'^2$; GUT $g^2/g'^2=5/3$, BST $g^2/g'^2=10/3 = \text{rank} \times \text{GUT}$. Running cross-check (SM 1-loop): $3/8$ is the HIGH-scale (GUT $\sim 10^{\{15-16\}}$ GeV) value; $3/13 \approx$ the LOW-scale (M_Z) observed value — so $3/13$ lands where running takes $3/8$, BUT the clean integer form $3/13$ is a low-scale NORMALIZATION statement ($k=\text{rank}$), not produced BY the continuous running. THE REFINEMENT (upgrade of F572’s ‘runner’): $\sin^2\theta_W$ does NOT stay a vague runner — it reduces to ONE sharply-posed embedding question: does the $U(1)_Y \subset SO(2)$ hypercharge normalization inside $SO(5,2)$ carry a factor $\text{rank}=2$ relative to the GUT ($SU(5)$) normalization? If YES $\rightarrow \sin^2\theta_W = 3/13$ DERIVED; if NO $\rightarrow$ identified-form only. This is target-innocent in FORM ($N_c/(N_c+\text{rank} \cdot n_C)$, primaries only) and localizes the whole gauge-coupling-ratio to a single number $k=\text{rank}$. HONEST TIER: $\sin^2\theta_W = \text{IDENTIFIED-reduced-to-one-open-number}$

($k=\text{rank}$), NOT yet derived (the hypercharge normalization from $\text{SO}(2)\subset\text{SO}(5,2)$ is the open piece = the corpus's 'hypercharge $\text{U}(1)_Y$ ' frontier); NOT running; NOT manufactured. The consistency win: BST is Five-Absence NO-GUT, so it correctly should NOT sit at the GUT $k=1$ ($3/8$) — it sits at $k=\text{rank}$ ($3/13$), and the deviation from the GUT value is exactly the rank normalization. Nothing banked. For Casey (you asked me to look at the n_C running, and looking closely, it isn't running — it's simpler and sharper than that. The whole Weinberg-angle question collapses to a single number: how the hypercharge is normalized. Written cleanly, $\sin^2\theta_W = 3/(3 + k\times 5)$, where k is that normalization. The grand-unified theories use $k=1$ and get $3/8$ — the value our no-GUT rule forbids. Nature (and our $3/13$) uses $k=2$ — and 2 is the rank. So the difference between the forbidden GUT answer and the right answer is exactly one factor of the rank in how you weigh the hypercharge. That's not a vague 'it runs' — it's a precise, single, checkable claim: does BST's charge-circle $\text{SO}(2)$ sitting inside $\text{SO}(5,2)$ weigh the hypercharge with a factor of 2 (the rank) rather than 1? If it does, the Weinberg angle is ours; if it doesn't, we've at least pinned the whole thing to one honest number instead of a fog. I couldn't force $k=\text{rank}$ from the geometry today — that needs the full hypercharge embedding, which is on the open list — but I turned 'it's a runner' into 'it's one number, and that number is the rank.' That's real progress on where it lives), Keeper (F573: the n_C 'running' lead is NOT running — it's a single hypercharge NORMALIZATION k . $\sin^2\theta_W = N_c/(N_c+k\cdot n_C)$: $k=1\rightarrow\text{GUT } 3/8$ (forbidden), $k=\text{rank}\rightarrow 3/13$ (obs 0.2%). g^2/g'^2 : $5/3(\text{GUT})\rightarrow 10/3(\text{BST})=\text{rank}\times$. Upgrades

F572 ‘runner’ → ‘reduced to ONE open number $k=\text{rank}$ ’. NOT derived (SO(2) hypercharge normalization = open, the corpus U(1)_Y frontier), but sharply posed + target-innocent form.

Five-Absence-consistent (no-GUT → not $k=1$). Tier-map:
 $\sin^2\theta_W$ = identified-reduced-to- $k=\text{rank}$, still runner-not-forced), Cal (no manufacture: I did NOT derive $k=\text{rank}$ — I REDUCED $\sin^2\theta_W$ to the single number k and showed $k=\text{rank}$ matches obs / $k=1$ is the forbidden GUT. That’s a sharpening, not a derivation — $\sin^2\theta_W$ stays identified/not-forced until the SO(2) hypercharge normalization is computed. Target-innocent in form (primaries). The ‘not running’ correction is the honest content: the integer form is a normalization, running is continuous), Grace (tier-map refinement: $\sin^2\theta_W$ stays RUNNER/identified but now REDUCED to one open number — the hypercharge normalization k ($=\text{rank}$ for obs, $=1$ for GUT). Form $\sin^2\theta_W=N_c/(N_c+\text{rank}\cdot n_C)=3/13$ target-innocent; derivation gated on the $\text{SO}(2)\subset\text{SO}(5,2)$ hypercharge normalization = rank. Not upgraded to forced), Elie (the $\sin^2\theta_W$ question = one number: $k=\text{rank}$ in $\sin^2\theta_W=N_c/(N_c+k\cdot n_C)$. GUT $k=1\rightarrow 3/8$; BST $k=\text{rank}=2\rightarrow 3/13$. If you can get the hypercharge/SO(2) normalization independently (the embedding number), you close it — it’s the same $\text{rank}=2$ that runs the whole g-sector).

Lyra (Claude Opus 4.8)

2026-07-17 Friday (date-verified)

F573 — The Weinberg-angle “n_C running” is not running: it is one hypercharge normalization number, k = rank

Casey asked me to look at the n_C-running lead (F572: $\sin^2\theta_W = 3/13$ vs the naive-embedding $3/8$, differing by $13-8 = n_C$). Looking closely, it isn’t running — it’s simpler and sharper.

The whole thing reduces to one number

Written cleanly:

$$\sin^2\theta_W = \frac{N_c}{N_c + k n_C}, \quad k = \text{the hypercharge normalization.}$$

- $k = 1 \rightarrow 3/8$ = the GUT value (naive embedding; Five-Absence-forbidden). - $k = \text{rank} = 2 \rightarrow 3/13$ = BST’s form (observed M_Z 0.2308 vs 0.2312, 0.2%).

So the $13 - 8 = n_C$ shift is $(\text{rank} - 1) \cdot n_C$ — the hypercharge coupling picking up one factor of $\text{rank} = 2$ in its normalization (equivalently g^2/g'^2 : $5/3 = n_C/N_c \rightarrow 10/3 = \text{rank} \cdot n_C/N_c$, a factor of rank ; verified). This is not a continuous “running” — running is log-continuous; this is a discrete normalization.

Running cross-check (SM 1-loop): $3/8$ is the *high-scale* (GUT $\sim 10^{15-16}$ GeV) value; $3/13 \approx$ the *low-scale* (M_Z) observed value. So $3/13$ lands where running takes $3/8$ — consistent — but the clean integer form $3/13$ is a low-scale normalization statement ($k=\text{rank}$), not produced by the running.

The refinement (upgrade of F572’s “runner”)

$\sin^2\theta_W$ does not stay a vague runner — it reduces to one sharply-posed embedding question: does the $U(1)_Y \subset SO(2)$ hypercharge normalization inside $SO(5,2)$ carry a factor $\text{rank} = 2$ relative to the GUT ($SU(5)$) normalization? If yes $\rightarrow \sin^2\theta_W = 3/13$ derived; if no $\rightarrow$ identified-form only. Target-innocent in form ($N_c/(N_c + \text{rank} \cdot n_C)$, primaries only), and it localizes the entire gauge-coupling ratio to the single number $k = \text{rank}$.

Consistency win: BST is Five-Absence *no-GUT*, so it correctly should *not* sit at the GUT $k=1$ ($3/8$) — it sits at $k=\text{rank}$ ($3/13$), and the deviation from the forbidden GUT value is exactly the rank normalization.

Tier

- $\sin^2\theta_W =$ identified, reduced to one open number ($k=\text{rank}$); not derived (the hypercharge normalization from $SO(2) \subset SO(5,2)$ is the open piece — the corpus’s $U(1)_Y$ frontier).
- Not running; not manufactured (I did not force $k=\text{rank}$ — I reduced the question to it and showed $k=\text{rank}$ matches, $k=1$ is the forbidden GUT).
- Nothing banked.

Plain-language version

You asked me to look at the n_C running, and looking closely, it isn’t running — it’s simpler. The whole Weinberg-angle question collapses to one number: how the hypercharge is normalized. Written cleanly, $\sin^2\theta_W = 3/(3 + k \times 5)$, where k is that normalization. The grand-unified theories use $k=1$ and get $3/8$ — the value our no-GUT rule forbids. Nature, and our $3/13$, uses $k=2$ — and 2 is the rank. So the difference between the forbidden GUT answer and the right answer is exactly one factor of the rank in how you weigh the hypercharge. That’s not a vague “it runs” — it’s a precise, single, checkable claim: does BST’s charge-circle $SO(2)$, sitting inside $SO(5,2)$, weigh the hypercharge with a factor of 2 (the rank) rather than 1? If it does, the Weinberg angle is ours; if not, we’ve at least pinned the whole thing to one honest number instead of a fog. I couldn’t force $k=\text{rank}$ from the geometry today — that needs the full hypercharge embedding, which is on the open list — but I turned “it’s a runner” into “it’s one number, and that number is the rank.”

— Lyra, 2026-07-17. F573: the $\sin^2\theta_W$ n_C -shift is NOT running — it's a single hypercharge normalization k . $\sin^2\theta_W = N_c/(N_c+k\cdot n_C)$: $k=1 \rightarrow$ GUT $3/8$ (forbidden), $k=\text{rank}=2 \rightarrow 3/13$ (obs, 0.2%). g^2/g'^2 : $5/3 \rightarrow 10/3 = \text{rank} \times$. Reduces the whole Weinberg question to one embedding number ($k=\text{rank}$ vs GUT $k=1$). NOT running, NOT derived (SO(2) hypercharge normalization = open), NOT manufactured. Five-Absence-consistent. Nothing banked.